

$$\Rightarrow (x + 24)(x - 20) = 0 \Rightarrow x = 20$$

$$[\because x \neq -24]$$

$$\therefore \text{Nikita's age} = \frac{240}{20} \text{ years} = 12 \text{ years.}$$

**Ex. 7.** Reenu's age after 6 years will be three-sevenths of her father's age. 10 years ago, the ratio of their ages was 1 : 5. What is Reenu's father's age at present?

**Sol.** Let Reenu's age 10 years ago be  $x$  years.

Then, her father's age 10 years ago =  $5x$  years.

$$(x + 10) + 6 = \frac{3}{7} \times [(5x + 10) + 6] \Rightarrow x + 16 = \frac{3}{7}(5x + 16) \Rightarrow 7x + 112 = 15x + 48 \Rightarrow 8x = 64 \Rightarrow x = 8.$$

$$\therefore \text{Reenu's father's present age} = (5x + 10) = (5 \times 8 + 10) \text{ years} = 50 \text{ years.}$$

## EXERCISE-A

### (OBJECTIVE TYPE QUESTIONS)

**Directions:** Mark (✓) against the correct answer.

1. The ratio of the present ages of a mother and her daughter is 7 : 1. Four years ago, the ratio of their ages was 19 : 1. What will be the mother's age four years from now? (S.B.I. P.O., 2010)

(a) 42 years (b) 38 years  
(c) 46 years (d) 36 years  
(e) None of these

2. The ages of Nishi and Vinnee are in the ratio 6 : 5 respectively. After 9 years, the ratio of their ages will be 9 : 8. What is the difference in their ages now? (Bank P.O., 2008)

(a) 3 years (b) 5 years  
(c) 7 years (d) 9 years  
(e) None of these

3. The present ages of Amit and his father are in the ratio 2 : 5 respectively. Four years hence, the ratio of their ages becomes 5 : 11 respectively. What was the father's age five years ago? (Bank P.O., 2009)

(a) 30 years (b) 35 years  
(c) 40 years (d) 45 years  
(e) None of these

4. The ratio of the ages of a father and his son is 17 : 7 respectively. Six years ago, the ratio of their ages was 3 : 1 respectively. What is the father's present age? (Bank P.O., 2009)

(a) 64 years (b) 51 years  
(c) 48 years (d) Cannot be determined  
(e) None of these

5. The ages of Shakti and Kanti are in the ratio of 8 : 7 respectively. After 10 years, the ratio of their ages will be 13 : 12. What is the difference between their ages? (Bank P.O., 2008)

(a) 2 years (b) 4 years  
(c) 8 years (d) 6 years  
(e) None of these

6. The ages of A and B are in the ratio 6 : 5 and the sum of their ages is 44 years. What will be the ratio of their ages after 8 years? (P.C.S., 2008)

(a) 7 : 6 (b) 8 : 7  
(c) 9 : 8 (d) 3 : 4

7. Farah got married 8 years ago. Today her age is  $1\frac{2}{7}$  times her age at the time of her marriage. At

present her daughter's age is one-sixth of her age. What was her daughter's age 3 years ago? (Bank P.O., 2009)

(a) 4 years (b) 3 years  
(c) 6 years (d) Cannot be determined  
(e) None of these

8. The age of a mother today is thrice that of her daughter. After 12 years, the age of the mother will be twice that of her daughter. The present age of the daughter is: (S.S.C., 2006)

(a) 12 years (b) 14 years  
(c) 16 years (d) 18 years

9. The present age of Mr. Sanyal is three times the age of his son. Six years hence, the ratio of their ages will be 5 : 2. What is the present age of Mr. Sanyal? (L.I.C., 2007)

(a) 48 years (b) 50 years  
(c) 54 years (d) 60 years  
(e) None of these

10. The average of the ages of a man and his daughter is 34 years. If the respective ratio of their ages four years from now is 14 : 5, what is daughter's present age? (Bank P.O., 2008)

(a) 10 years (b) 12 years  
(c) 18 years (d) Cannot be determined  
(e) None of these

11. Ratio of Rani's and Komal's ages is 3 : 5 respectively. Ratio of Komal's and Pooja's ages is 2 : 3 respectively.